

Speculative Self-Distillation enables Efficient Knowledge Internalization

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TL;DR

Problem. Self-distillation teaches a model a document by training it to match itself with the document in context. **on-policy** rollouts are accurate but spend most tokens where the teacher adds little.

Method. **SSD**: the student writes the rollout; the teacher writes only **inflection tokens**, where the two models disagree by more than τ .

Result. **SSD** matches **on-policy** accuracy with 45% fewer supervised tokens on three tasks and **forgets** less.

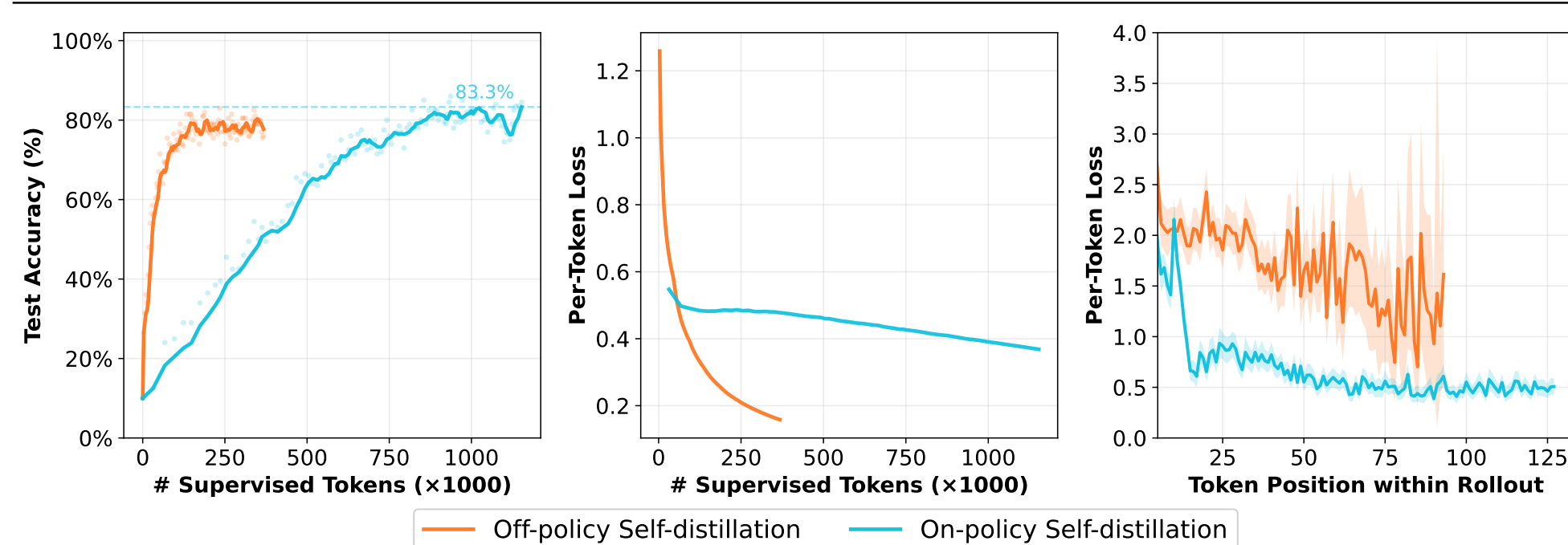
Self-Distillation

- Teacher $\pi_T(\cdot | x, C)$: the model with privileged context C (a document). Student $\pi_\theta(\cdot | x)$: the same model without it, trained to no longer need C .

$$\mathcal{L}(\theta) = \mathbb{E}_{y \sim \pi_{\text{gen}}} \left[\frac{1}{|y|} \sum_t D(\pi_T(\cdot | x, C, y_{<t}) \| \pi_\theta(\cdot | x, y_{<t})) \right]$$

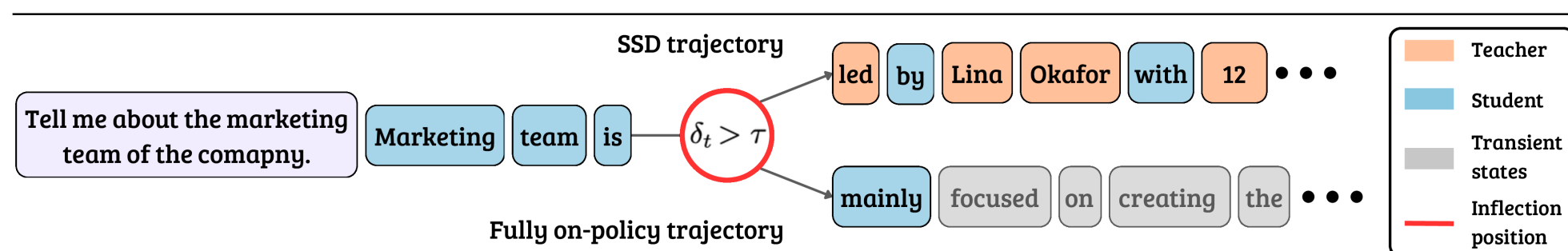
- Methods differ only in the rollout policy π_{gen} :
 - off-policy** ($\pi_{\text{gen}} = \pi_T$): strong signal, but trains on states the student never visits.
 - on-policy** ($\pi_{\text{gen}} = \pi_\theta$, SDFT): no train-test mismatch, but slow.
- SSD** uses a **per-token mix** of the two.

Why On-Policy Is Slow



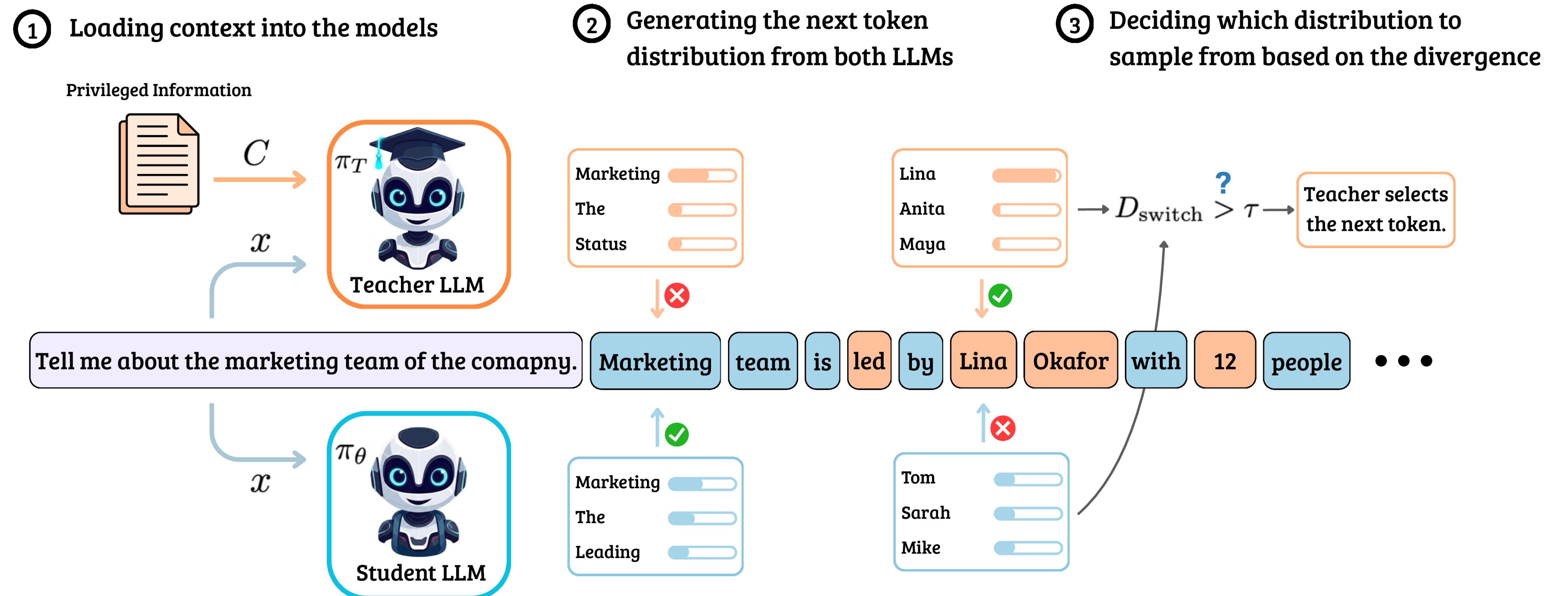
- off-policy** learns $\sim 5\times$ faster but plateaus lower.
- Along **on-policy** rollouts the per-token signal **collapses after a few tokens**.

Inflection Tokens



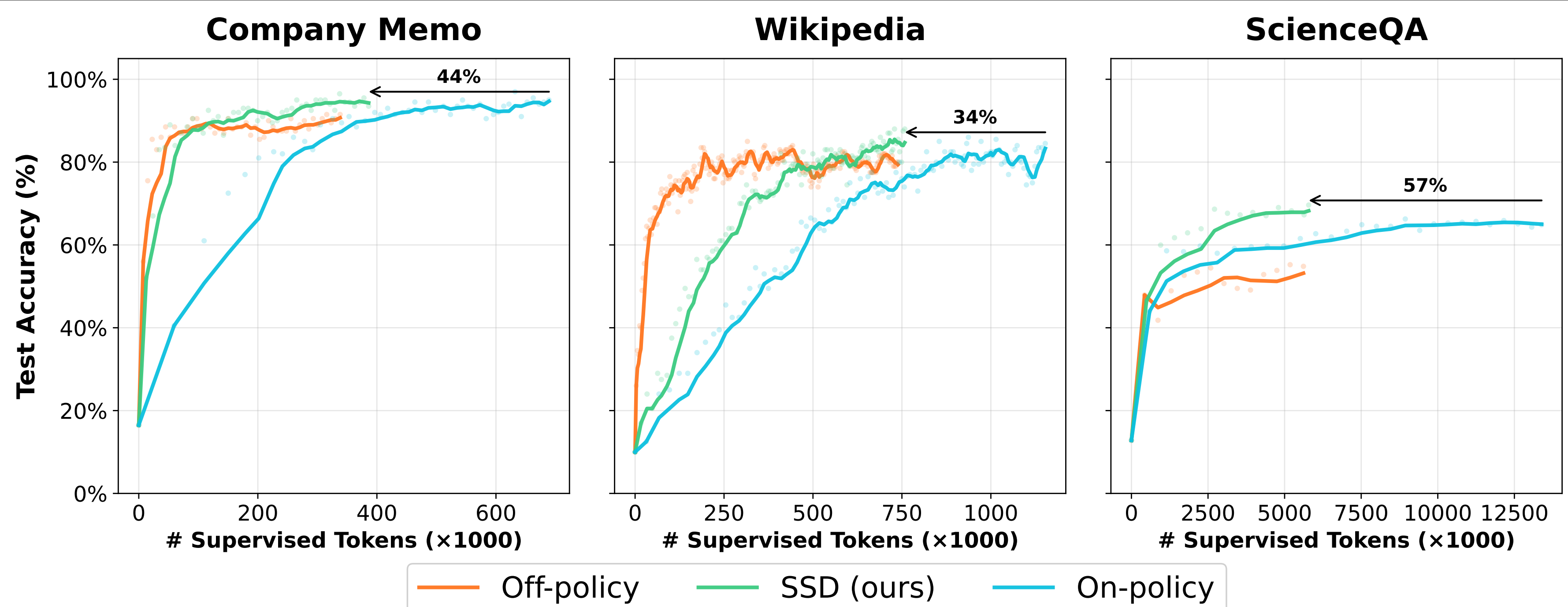
- Where the document changes the next token ($\delta_t > \tau$), the teacher writes.
- on-policy** keeps the student's token and drifts into **transient states**.

Speculative Self-Distillation (SSD)



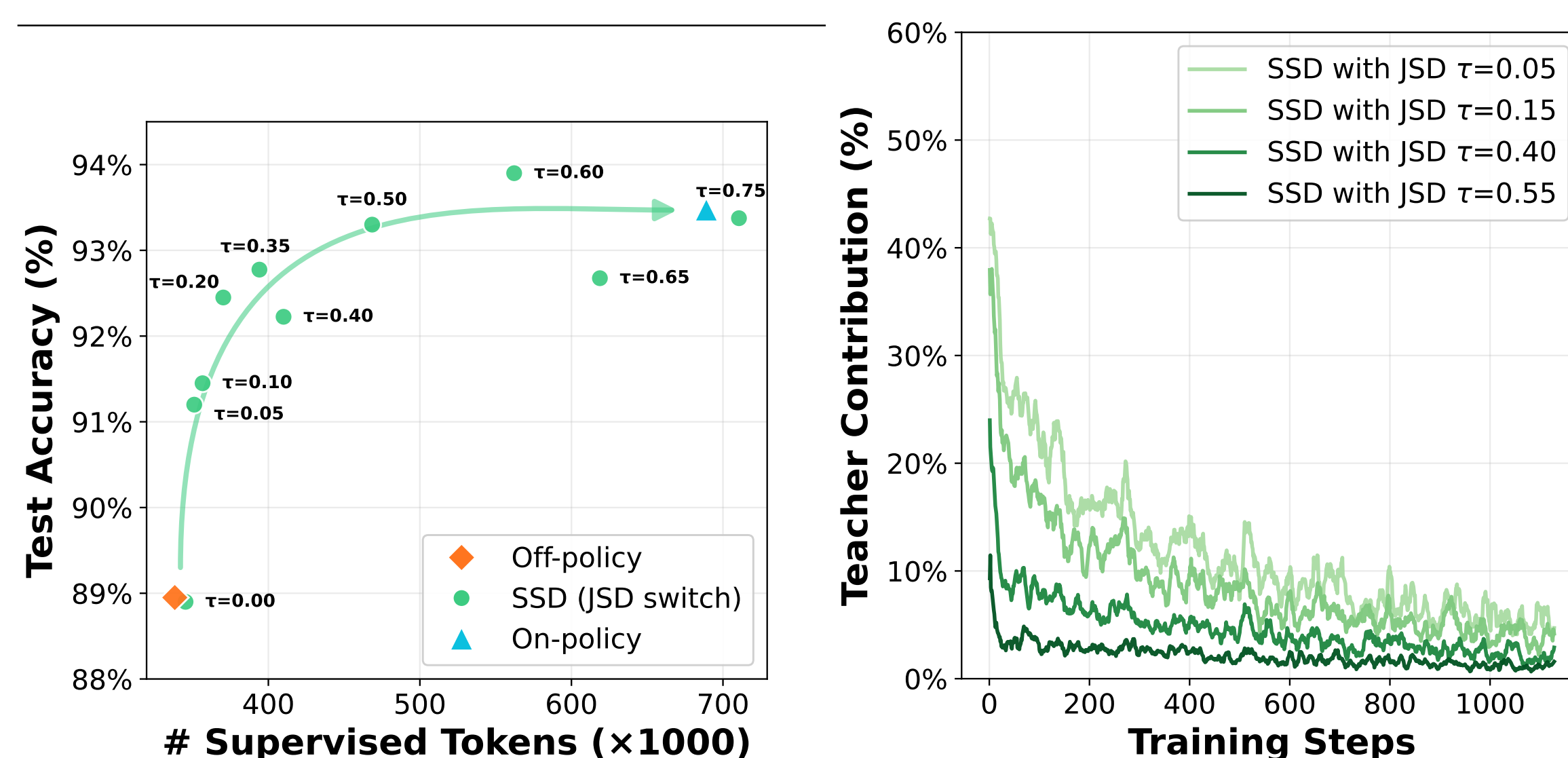
- Student generates by default.
- If $D_{\text{switch}}(\pi_\theta \| \pi_T) > \tau$, the teacher picks the token.
- Every position is trained with the same loss.
- $\tau \rightarrow 0$ is **off-policy**, $\tau \rightarrow \infty$ is **on-policy**.

SSD Reaches On-Policy Accuracy with 34–57% Fewer Tokens



Qwen3-4B-Instruct as student and frozen teacher on three closed-book QA tasks: a synthetic company memo, a post-cutoff Wikipedia article, and SciKnowEval chemistry. Cost is supervised tokens.

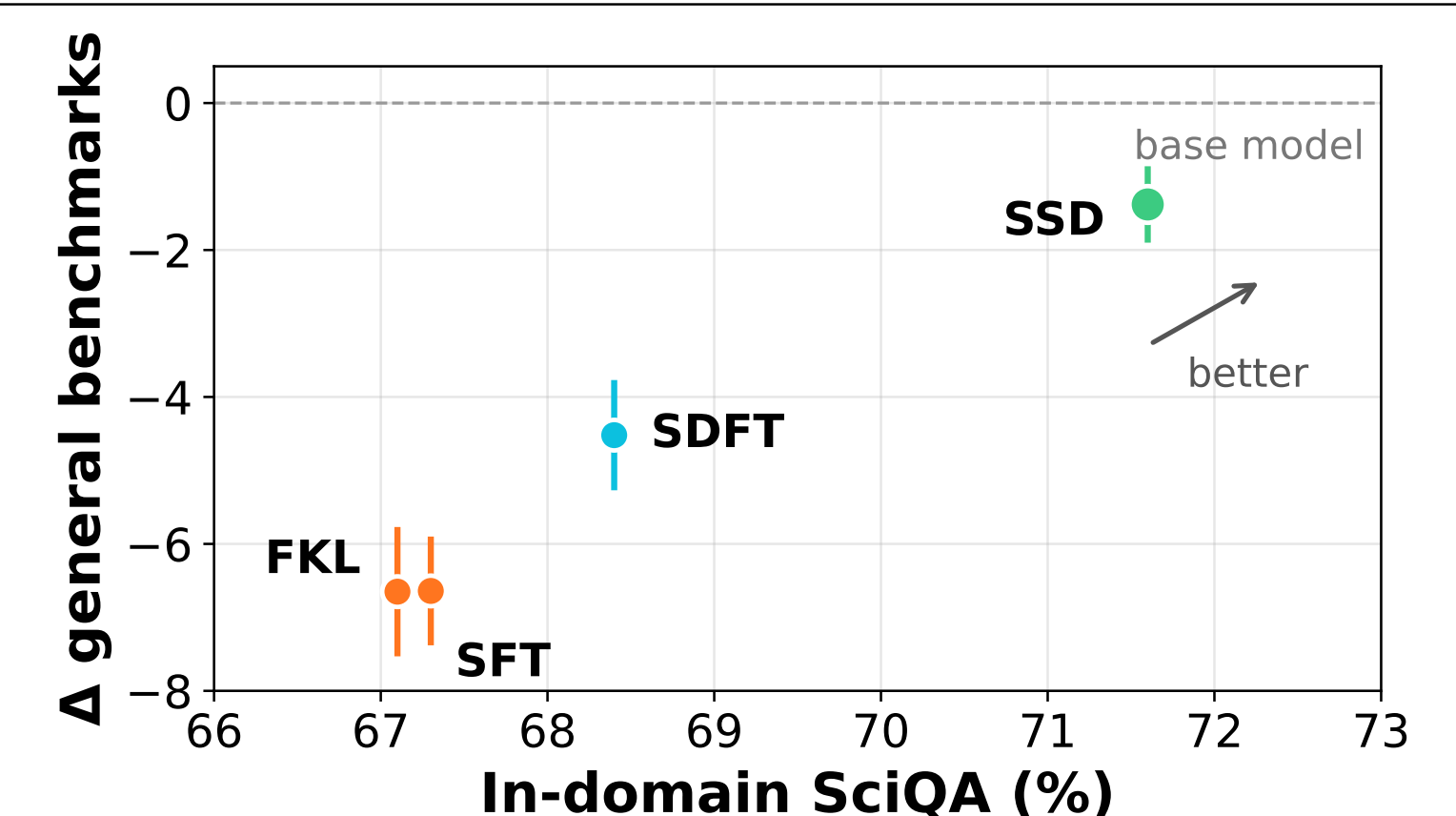
One Knob, and a Built-in Curriculum



$\tau \in [0.2, 0.5]$ sits on the Pareto frontier.

The teacher's share fades with no schedule.

Less Forgetting



Qwen2.5-7B, SciKnowEval chemistry: best in-domain accuracy, smallest drop on six general benchmarks.